Synopsis

ONLINE PURCHASE PREDICTOR

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- **Problem Description:**

With the emergence of e-commerce platforms, understanding customer behaviour has become very important for businesses. Many users visit online shopping websites, browse products, and view different pages, but only a small percentage of them actually end up buying something online. Predicting whether a customer will buy a product based on their browsing behaviour can help companies improve their marketing strategies, personalize recommendations, and increase sales.

The problem statement states to develop a machine learning classification system that makes predictions of whether the user makes a purchase or not based on their online browsing behavior. The dataset includes behavioral attributes such as browsing time, number of clicks, product views, session duration, and page visits. These features provide useful information about the level of customer interest in a product.

Before building the predictive model, the dataset needs to be preprocessed. This includes handling missing values, removing duplicate or irrelevant records, and applying feature scaling so that all features are in a similar range. Feature selection techniques are used to eliminate low-impact attributes and keep only the most important behavioural features that influence purchasing decisions.

After preprocessing, classification algorithms such as Logistic Regression and K-Nearest Neighbors (KNN) are applied to train the model. The performance of the model is evaluated using accuracy, precision, and F1-score to ensure reliable prediction of customer purchasing behaviour.

- **Steps of Implementation:**

1. Data Collection:

The first phase involves gathering the dataset containing customer behavioral attributes. This includes tracking metrics such as:

Browsing time and session duration.

Number of clicks and product views.

Specific page visits during a shopping session.

2. Data Preprocessing:

Before the model can learn, the raw data must be cleaned and organized. Key tasks include:

- Cleaning: Handling missing values and removing duplicate or irrelevant records.
- Feature Scaling: Adjusting features so they reside within a similar range for better mathematical processing.

- Feature Selection: Using techniques to eliminate low-impact attributes and retain only the most influential variables

3. Model Selection:

In this phase, appropriate classification algorithms are chosen to handle the binary task (Purchase vs. No Purchase). The primary algorithms identified for this project are:

- Logistic Regression.
- K-Nearest Neighbors (KNN).

4. Training and Validation

The preprocessed data is fed into the selected algorithms. The model learns to associate specific browsing patterns (like high click counts) with the likelihood of a purchase. Validation is performed to ensure the model isn't just "memorizing" the data but actually learning patterns

5. Evaluation

The model's performance is tested using unseen data to ensure reliability. Success is measured using:

Accuracy, Precision, and F1-score to determine how well it predicts customer behavior.

6. Final Model

Once the performance meets the required standards, the Final Model is generated. This model can then be used by e-commerce platforms to personalize recommendations and improve marketing strategies in real-time.

Expected Modules/ Libraries to be used :

1. **NumPy:**

For numerical computations and handling arrays.

2. **Pandas:**

For data manipulation, cleaning, and analysis.

3. **Matplotlib:**

For basic data visualization (graphs and plots).

4. **Seaborn:**

For advanced and attractive statistical visualizations.

5. **Scikit-learn (sklearn):**

For machine learning tasks:

- Data preprocessing (scaling, encoding):
- Feature selection
- Algorithms like Logistic Regression and KNN
- Model evaluation (accuracy, precision, F1-score)

6. **SciPy (optional):**

For additional scientific and statistical computations.

7. **Warnings / OS (optional):**

For handling system operations and suppressing unnecessary warnings.

- **Expected Evaluation metrics and Plots to be used:**

To evaluate the performance of the proposed machine learning model for predicting online purchase behavior, several evaluation metrics and visualization techniques will be used. Since the problem is a binary classification task (purchase or no purchase), multiple metrics are considered to ensure a comprehensive assessment of the model.

- **Required topics from the subject:**

1. Data Preprocessing:

Data preprocessing involves cleaning and transforming raw data into a usable format. It includes handling missing values, removing duplicates, and scaling features. This step improves model accuracy and performance.

2. Feature Selection:

Feature selection is the process of choosing the most relevant features for model training. It removes unnecessary or low-impact variables. This helps reduce complexity and improves model efficiency.

3. Logistic Regression:

Logistic Regression is a supervised learning algorithm used for classification problems. It predicts the probability of a binary outcome using a sigmoid function. It is simple and works well for linearly separable data.

4. K-Nearest Neighbors (KNN):

KNN is a supervised learning algorithm that classifies data based on the nearest neighbors. It calculates distance between data points to make predictions. It is useful for both linear and non-linear data patterns.

5. Model Evaluation Metrics:

Model evaluation metrics like accuracy, precision, and F1-score measure the performance of a model. Accuracy shows overall correctness, while precision and F1-score give better insight for imbalanced data. These metrics help in selecting the best model.

6. Feature Scaling:

Feature scaling ensures that all features have similar ranges. Techniques like Min-Max scaling and standardization are used. It is important for algorithms like KNN and Logistic Regression to perform effectively.

Platform Required:

Jupyter Notebook

- **Books and Link Sources:**
  1. Fundamental of AI and ML Textbook
  2. Scikit Learn Documentation: scikit-learn.org
  3. Data Preprocessing and Feature Engineering for Data Mining: www.researchgate.net
  4. GeeksforGeeks (for reading up various topics of machine learning) : www.geeksforgeeks.org